```
pip install tf-Keras
```

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Requirement already satisfied: tf-Keras in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: tensorflow<2.21,>=2.20 in /usr/local/lib/python
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/loc
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python3.12/
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/pyth
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.
Requirement already satisfied: tensorboard~=2.20.0 in /usr/local/lib/python3.
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist
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Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyth
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/di
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pytho
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-pa
```

```
from sentence_transformers import SentenceTransformer,util
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]
```

```
s1 = "This  is on the mat"
s2 = "A cat sitting on a mat"
```

```
emb = model.encode(s1)
emb = model.encode(s2)

sentences = [
    "The cat sits outside",
    "A man is playing guitar",
    "I love pasta",
    "The new movie is awesome",
    "The cat plays in the garden",
    "A woman watches TV",
    "The new movie is so great",
    "Do you like pizza?",
    "She enjoys reading books",
    "He is walking his dog",
    "The weather is nice today",
    "Let's go to the beach",
    "I need a cup of coffee",
    "They are planning a trip",
    "She cooks delicious meals",
    "I am learning to play the piano",
    "He works at a tech company",
    "This book is very interesting",
    "The sun is shining brightly",
    "It's raining outside",
    "The children are playing soccer",
    "He is fixing the car",
    "She writes beautiful poetry",
    "The flowers are blooming",
    "I enjoy hiking in the mountains",
    "He is swimming in the pool",
    "We are watching a comedy show",
    "The dog is barking loudly",
    "She is painting a landscape",
    "He likes to play video games",
    "I am studying for my exams",
    "They are building a new house",
    "The coffee shop is busy",
    "He listens to classical music",
    "She is designing a website",
    "I bought a new laptop",
    "They are baking cookies",
    "The bird sings in the morning",
    "He loves going to concerts",
    "She practices yoga every day",
    "The phone is ringing",
    "I need to finish this project",
    "The restaurant has great food",
    "He collects rare coins",
    "She enjoys watching documentaries",
```

```
    "They are exploring the city",
    "The laptop is charging",
    "He is training for a marathon",
    "I meditate every morning",
    "The bakery sells fresh bread",
    "She is learning French"
]

paraphrases = util.paraphrase_mining(model, sentences)
paraphrases[:20]
```

```
[[0.8939038515090942, 3, 6],
 [0.6787886619567871, 0, 4],
 [0.6313821077346802, 8, 44],
 [0.5843963623046875, 13, 45],
 [0.5794311761856079, 33, 38],
 [0.5284280180931091, 12, 32],
 [0.5188640356063843, 29, 38],
 [0.5187947750091553, 35, 46],
 [0.5095502138137817, 2, 7],
 [0.5038483142852783, 8, 50],
 [0.5013786554336548, 14, 42],
 [0.48317253589630127, 29, 33],
 [0.4814086854457855, 15, 30],
 [0.478863000869751, 22, 28],
 [0.4751392900943756, 10, 18],
 [0.4729399085044861, 5, 44],
 [0.47274249792099, 14, 22],
 [0.4725485146045685, 28, 34],
 [0.46822240948677063, 36, 49],
 [0.4525766968727112, 8, 22]]
```

```
for i in paraphrases[0:20]:
  score,i,j=i
  print("{} \t\t {} \t\t Score: {:.4f}".format(sentences[i],sentences[j],scor
```

```
The new movie is awesome         The new movie is so great       Score: 0.893
The cat sits outside         The cat plays in the garden         Score: 0.678
She enjoys reading books         She enjoys watching documentaries       Scor
They are planning a trip         They are exploring the city         Score: 0
He listens to classical music        He loves going to concerts          Scor
I need a cup of coffee       The coffee shop is busy         Score: 0.5284
He likes to play video games         He loves going to concerts          Scor
I bought a new laptop        The laptop is charging          Score: 0.5188
I love pasta         Do you like pizza?          Score: 0.5096
She enjoys reading books         She is learning French          Score: 0.503
She cooks delicious meals        The restaurant has great food       Score: 0
He likes to play video games         He listens to classical music       Scor
I am learning to play the piano          I am studying for my exams          
```

```
She writes beautiful poetry          She is painting a landscape         Scor
The weather is nice today        The sun is shining brightly        Score: 0
A woman watches TV       She enjoys watching documentaries       Score: 0.472
She cooks delicious meals        She writes beautiful poetry        Score: 0
She is painting a landscape          She is designing a website          Scor
They are baking cookies          The bakery sells fresh bread        Score: 0
She enjoys reading books         She writes beautiful poetry        Score: 0

model=SentenceTransformer('distiluse-base-multilingual-cased-v1')

modules.json:   0%|          | 0.00/341 [00:00<?, ?B/s]

config_sentence_transformers.json:   0%|          | 0.00/122 [00:00<?, ?B/s]

README.md:   0%|          | 0.00/2.24k [00:00<?, ?B/s]

sentence_bert_config.json:   0%|          | 0.00/53.0 [00:00<?, ?B/s]

config.json:   0%|          | 0.00/556 [00:00<?, ?B/s]

model.safetensors:   0%|          | 0.00/539M [00:00<?, ?B/s]

Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]

tokenizer_config.json:   0%|          | 0.00/452 [00:00<?, ?B/s]

vocab.txt:   0%|          | 0.00/996k [00:00<?, ?B/s]

tokenizer.json:   0%|          | 0.00/1.96M [00:00<?, ?B/s]
```

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special_tokens_map.json:   0%|          | 0.00/112 [00:00<?, ?B/s]



config.json:   0%|          | 0.00/190 [00:00<?, ?B/s]



config.json:   0%|          | 0.00/114 [00:00<?, ?B/s]



2_Dense/model.safetensors:   0%|          | 0.00/1.58M [00:00<?, ?B/s]

corpus1 = ["The stock market sawe a big drope",
           "sharest declined sharply sue to economic news.",
           "Mony gets money in stock",
           "It may rain today"]
corpus_emb=model.encode(corpus1)

# Copute paiwise Consine similarity
cos_sim=util.cos_sim(corpus_emb,corpus_emb)
avg_sim= cos_sim.mean(dim=1)
outlier_index=avg_sim.argmin().item()
outlier_index
print(f"outlier sentence:{corpus1[outlier_index]} (AVG Similarity:{avg_sim[ou

outlier sentence:It may rain today (AVG Similarity:0.33442366123199463)

corpus2 = ["She is giving a presentation at the conference tomorrow.",
           "The old bridge started giving under the weight of the heavy truck
           " A cats are sleeping on mat",
           "The phone is ringing"]
corpus_emb=model.encode(corpus2)

# Copute paiwise Consine similarity
cos_sim=util.cos_sim(corpus_emb,corpus_emb)
avg_sim= cos_sim.mean(dim=1)
outlier_index=avg_sim.argmin().item()
outlier_index
print(f"outlier sentence:{corpus2[outlier_index]} (AVG Similarity:{avg_sim[ou

outlier sentence:The old bridge started giving under the weight of the heavy t

corpus3 = ["sita is singing a song",
           "the boy playing a games ",
           "The dog is barking loudly"]
corpus_emb=model.encode(corpus3)
```

```
# Copute paiwise Consine similarity
cos_sim=util.cos_sim(corpus_emb,corpus_emb)
avg_sim= cos_sim.mean(dim=1)
outlier_index=avg_sim.argmin().item()
outlier_index
print(f"outlier sentence:{corpus3[outlier_index]} (AVG Similarity:{avg_sim[ou

outlier sentence:the boy playing a games  (AVG Similarity:0.41973280906677246
```